Portable Identity Framework: A Framework for Transferable Character Identity

Author: Najee Jamison

Abstract

Artificial characters are typically defined by the systems that execute them. A companion created for a chatbot, game, virtual environment, or robotic platform often cannot be transferred to another system without rebuilding significant portions of its personality model. Consequently, identity becomes dependent upon implementation.

The Portable Identity Framework (PIF) defines identity as a compiled, transferable identity that exists independently of any execution system. Identity is derived from a Character Bible, validated, compiled, and packaged into a portable form containing personality structures, values, relationships, behavioral tendencies, signature behaviors, and verification metadata. Execution systems interpret the portable identity through embodiment-specific runtimes while preserving a consistent underlying character.

Identity-Constrained Character Robotics (ACR) serves as the first implementation of this framework. Within ACR, the portable identity is represented by the Immutable Core Identity, demonstrating how a character can be authored once and instantiated within a robotic embodiment without altering its defining characteristics.

Introduction

Current character systems bind identity to implementation. Personality is commonly derived from prompts, model configurations, memory systems, application logic, fine-tuning procedures, or platform-specific behavior. When those underlying systems change, the character often changes as well.

Because identity is tied to implementation, moving a character between systems frequently requires reconstruction of its personality model. This makes long-term continuity difficult and limits portability across platforms.

The Portable Identity Framework addresses this limitation by treating identity as an independent element rather than a property of the system executing it. The framework allows a character to retain its defining traits while operating through multiple embodiments and execution environments.

Relationship to Identity-Constrained Character Robotics

Identity-Constrained Character Robotics (ACR) provides the first implementation of the Portable Identity Framework and serves as its reference architecture.

Within ACR, a Character Bible is compiled into an Immutable Core Identity, verified through integrity checks, and deployed through a controlled transfer process. The robot interprets the identity but does not modify it.

The Portable Identity Framework generalizes this architecture. Rather than limiting deployment to robots, the same portable identity may be interpreted by chatbots, game agents, virtual avatars, educational systems, and future AI platforms.

Within ACR, the portable identity is implemented as the Immutable Core Identity, compiled from a Character Bible, verified through checksum validation, and transferred through a controlled deployment process.

Identity Compilation

The framework defines identity as a compiled identity rather than a runtime configuration.

Identity begins as a Character Bible containing personality traits, values, relationships, developmental history, preferences, limitations, and signature behaviors. The Bible serves as the canonical source document for the character.

Before deployment, the Bible is validated for completeness and consistency. Validation ensures that the identity contains sufficient information to support interpretation across embodiments.

Compilation transforms the validated Bible into a portable identity consisting of structured behavioral representations. These representations may include preference distributions, value structures, relationship definitions, contextual tendencies, and behavioral signatures.

The resulting identity receives version information, verification metadata, and provenance records to support maintenance, validation, and transfer.

Portable Identity Structure

A portable identity contains the information necessary to preserve character continuity across embodiments.

Its contents include:

- Personality traits and dispositions
- Value structures and priorities
- Relationship definitions
- Behavioral tendencies
- Signature behaviors
- Version information
- Verification metadata
- Authorship records
- Provenance information

The portable identity defines the character itself. It does not contain embodiment-specific behaviors, hardware descriptions, rendering logic, motion systems, or platform-dependent code.

Execution and Embodiment

The framework separates identity from execution.

Identity defines a character's values, tendencies, preferences, and relationships.

Execution systems interpret those characteristics and convert them into decisions.

Embodiments provide the mechanisms through which those decisions are expressed.

A chatbot may express identity through language. A robot may express identity through movement and interaction. A game character may express identity through gameplay behavior. A virtual avatar may express identity through animation and social interaction.

Although embodiments differ in capability, each derives behavior from the same portable identity.

Identity Runtime

The framework requires an Identity Runtime.

The runtime loads, validates, interprets, and exposes the portable identity to an execution system. It serves as the interface between identity and embodiment.

A runtime may be connected to a language model, robot controller, game engine, simulation environment, or other execution system. Regardless of platform, the runtime preserves the portable identity as the authoritative source of character definition.

Identity Transfer

Identity transfer moves a portable identity from one execution environment to another.

The transfer process includes validation and verification steps that ensure the receiving system obtains the same identity that was originally compiled.

Because the identity remains unchanged, continuity is preserved across embodiments, hardware generations, software revisions, and execution platforms.

Framework Integration

The Portable Identity Framework is designed to operate alongside existing systems.

A chatbot, robot controller, game engine, or virtual world can retain its existing execution architecture while sourcing character definitions from a portable identity.

This allows identity to remain stable even as execution technologies evolve.

Applications

The framework may be applied to:

- Companion chatbots
- Humanoid robots
- Virtual companions
- Game characters
- Educational tutors
- Digital twins
- Augmented reality assistants
- Persistent AI agents

A portable identity allows the same character to appear across multiple embodiments while maintaining consistent values, tendencies, relationships, and behavioral signatures.

Conclusion

The Portable Identity Framework defines identity as a compiled and transferable identity independent of any particular execution system.

By separating identity from implementation, the framework allows a character to persist across multiple embodiments while maintaining continuity of personality, values, relationships, and behavior.

Identity-Constrained Character Robotics demonstrates the first implementation of this concept through immutable identities, verification mechanisms, and controlled deployment. The Portable Identity Framework extends the same principle beyond robotics, providing a foundation for characters that can persist across platforms, embodiments, and future generations of AI systems.